

SYDNEY TECHNICAL HIGH SCHOOL



MATHEMATICS EXTENSION 1

YEAR 11 PRELIMINARY HSC

ASSESSMENT TASK II

JULY 2011

General Instructions:

- Working time allowed – 70 minutes.
- Write using black or blue pen.
- Approved calculators may be used.
- All necessary working should be shown.
- Start each question on a new page.
- Attempt all questions.

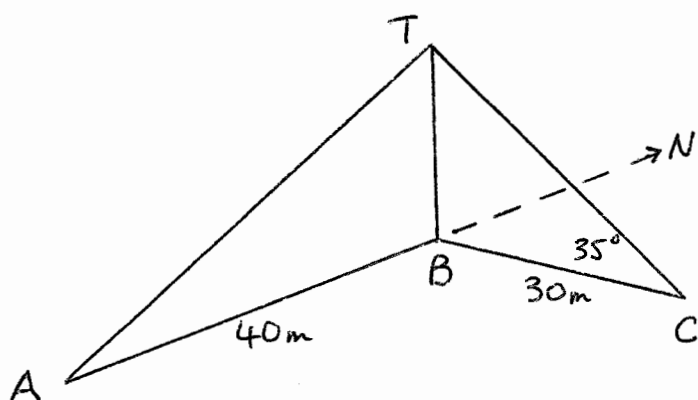
NAME: _____

TEACHER: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	TOTAL
/10	/9	/9	/9	/10	/10	/57

Question 1**Marks**

- a) i) Convert 40° to radians. 1
- ii) Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$ 2
- b) i) Write the expansion of $\cos(A + B)$. 1
- ii) Hence find the value of $\cos 75^\circ$ in simplest exact form. 2
- c)



Points A,B,C are on
level, horizontal ground.

A vertical tower BT is observed due north of A at a distance of 40 m. It is also observed from C, 30 m away and on a bearing of 050° from B, with an angle of elevation of 35° .

- i) Find BT. 1
- ii) Find $\angle BAT$ (to 1 dec. place). 1
- iii) Find AC (to 1 dec. place). 2

Question 2 (start a new page)

- a) Solve $\cos 2\theta = \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$. 3
- b) If $\sin \theta = 0.3$, where θ is acute, find without a calculator, the value of $\sin 2\theta$. 2

Leave your answer in exact form.

c) Use the “t results” to :

i) solve $\cos \theta - \sin \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$, 3

ii) find the exact value of $\frac{2 \tan 22.5^\circ}{1 + \tan^2 22.5^\circ}$ 1

Question 3 (start a new page)

a) Given that $3 \sin \theta - 2 \cos \theta = A \sin(\theta - \alpha)$,

i) find A and α . 2

ii) Hence, solve $3 \sin \theta - 2 \cos \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$. 2

(answer to 1 decimal place)

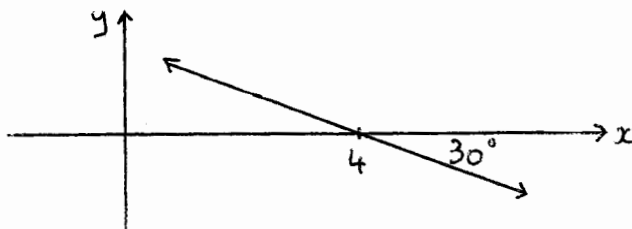
iii) Find the first positive θ such that $3 \sin \theta - 2 \cos \theta$ has a maximum value. 1

b) i) Write the expansion of $\tan 2\theta$. 1

ii) Hence, find the value of $\tan 22.5^\circ$ in simplest exact form. 3

Question 4 (start a new page)

a) Find the equation of this line: 2



b) Point P divides the interval A(-2, 3), B(6, -1) externally in the ratio 1:3.

i) Show the relative positions of A, B, P using the given ratio. There is no need to plot A and B on a number plane. 1

ii) Find the coordinates of P. 2

c) Find the acute angle, correct to 1 decimal place, between:

i) the lines $y = 3x + 2$ and $2x + 2y - 5 = 0$, 2

ii) the line $y = 3x + 2$ and the y axis. 2

Question 5

- a) Given $A(0,0)$, $B(4,-2)$, $C(3,3)$ and $D(9,7)$. P is the midpoint of AB and Q is the midpoint of CD.
- i) Find the equation of the perpendicular bisector of PQ. 3
Give your answer in general form.
- ii) Find the coordinates of a point E such that ABCE is a parallelogram. 1
- b) Differentiate: i) $y = \frac{1}{\sqrt{4x-2}}$ 2
ii) $y = \frac{2x+1}{(4x-9)^4}$ Leave your answer fully simplified. 3
- c) Evaluate $\lim_{x \rightarrow \infty} \frac{x^2-5x+2}{2x^2+3x}$ 1

Question 6 (start a new page)

- a) Explain the essential geometrical distinction between $\frac{f(x+h)-f(x)}{h}$ 1
and $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$
- b) Use first principles to find the derivative of $y = \frac{1}{x^2}$. 3
- c) Find the equation of the normal to the curve $y = \frac{1}{2x}$ at the point where $x = 1$. 3
- d) Find the values of k such that the line $3x - 4y + k = 0$ intersects twice with the circle $(x - 4)^2 + y^2 = 4$. 3

END OF TEST

SOLUTIONS

① a) i) $\frac{40\pi}{180} = \underline{\underline{\frac{2\pi}{9}}}$ ii) $\theta = 60^\circ$ (3rd, 4th quadrants)
 $= 240^\circ, 300^\circ$
 $= \underline{\underline{\frac{4\pi}{3}, \frac{5\pi}{3}}}$

b) i) $\cos A \cos B - \sin A \sin B$

ii) $\cos 75 = \cos(45 + 30)$
 $= \cos 45 \cos 30 - \sin 45 \sin 30$
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \times \frac{1}{2}$
 $= \underline{\underline{\frac{\sqrt{3}-1}{2\sqrt{2}}}}$

c) i) $\tan 35 = \frac{BT}{30}$

$$BT = 30 \tan 35$$
$$= \underline{\underline{21}}$$

ii) $\tan A = \frac{30 \tan 35}{40}$

$$\therefore \underline{\underline{A = 27.7^\circ}}$$

iii) $AC^2 = 40^2 + 30^2 - 2 \times 40 \times 30 \times \cos 130^\circ$
 $= 4042.7$

$$\therefore \underline{\underline{AC = 63.6 \text{ m.}}}$$

② a) $2\cos^2\theta - 1 = \cos\theta$

$$2\cos^2\theta - \cos\theta - 1 = 0$$

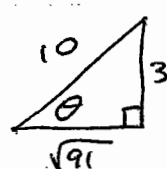
$$(2\cos\theta + 1)(\cos\theta - 1) = 0$$

$$\cos\theta = -\frac{1}{2} \text{ or } 1$$

$$\theta = 60^\circ \text{ (2nd, 3rd quadrants), } 0^\circ, 360^\circ$$

$$= \underline{\underline{120^\circ, 240^\circ, 0^\circ, 360^\circ}}$$

b) $\sin 2\theta = 2\sin\theta \cos\theta$



$$= 2 \times \frac{3}{10} \times \frac{\sqrt{91}}{10}$$

$$= \frac{6\sqrt{91}}{100}$$

$$\text{or } \frac{3\sqrt{91}}{50}$$

$$\text{or } 0.6\sqrt{10}$$

$$c) i) \frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} = 1$$

$$\therefore \cancel{1-t^2} - 2t = \cancel{1+t^2}$$

$$\therefore -2t^2 - 2t = 0$$

$$-2t(t+1) = 0$$

$$t = 0 \text{ or } -1$$

$$\tan \frac{\theta}{2} = 0 \text{ or } -1$$

$$\frac{\theta}{2} = 0^\circ, 180^\circ, \cancel{360^\circ}, 135^\circ, \cancel{315^\circ}$$

$$\therefore \underline{\underline{\theta = 0^\circ, 360^\circ, 270^\circ}}$$

$$ii) \frac{2t}{1+t^2} = \sin \theta \quad \left(t = \tan \frac{45^\circ}{2} \right)$$

$$\theta = 45^\circ$$

$$= \sin 45^\circ$$

$$= \underline{\underline{\frac{1}{\sqrt{2}}}}$$

$$③ a) i) \underline{A = \sqrt{13}}$$

$$3 \sin \theta - 2 \cos \theta = \sqrt{13} \sin(\theta - \alpha)$$

$$\frac{3}{\sqrt{13}} \sin \theta - \frac{2}{\sqrt{13}} \cos \theta = \sin(\theta - \alpha)$$

$$= \sin \theta \cos \alpha - \cos \theta \sin \alpha$$

$$\therefore \left. \begin{aligned} \cos \alpha &= \frac{3}{\sqrt{13}} \\ \sin \alpha &= \frac{2}{\sqrt{13}} \end{aligned} \right\} \begin{aligned} &\text{1st quad.} \\ &\underline{\underline{\alpha = 33.7^\circ}} \end{aligned}$$

$$ii) \sqrt{13} \sin(\theta - 33.7^\circ) = 1$$

$$\sin(\theta - 33.7^\circ) = \frac{1}{\sqrt{13}}$$

$$\therefore \theta - 33.7 = 16.1^\circ \text{ (1st, 2nd quad.)}$$

$$= 16.1^\circ \text{ or } 163.9^\circ$$

$$\therefore \underline{\underline{\theta = 49.8^\circ \text{ or } 197.6^\circ}}$$

$$b) i) \tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$ii) \tan 45^\circ = \frac{2 \tan 22.5^\circ}{1 - \tan^2 22.5^\circ}$$

$$1 - \tan^2 22.5 = 2 \tan 22.5$$

$$\tan^2 22.5 + 2 \tan 22.5 - 1 = 0$$

$$\tan 22.5 = \frac{-2 \pm \sqrt{4+4}}{2}$$

$$= \frac{-2 \pm 2\sqrt{2}}{2}$$

$$= -1 \pm \sqrt{2}$$

$$\text{But } \tan(\text{acute}) > 0$$

$$\therefore \underline{\underline{\tan 22.5^\circ = -1 + \sqrt{2}}}$$

$$iii) \text{Max. value is when } \sin(\theta - 33.7^\circ) = 1$$

$$\therefore \theta - 33.7^\circ = 90^\circ$$

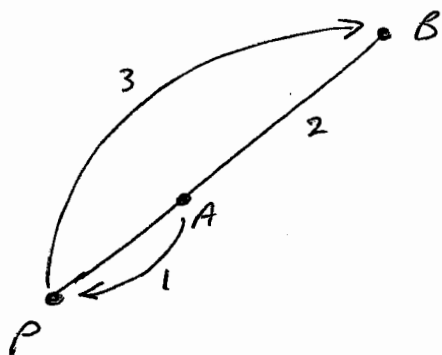
$$\therefore \underline{\underline{\theta = 123.7^\circ}}$$

4) a) $m = \tan 150^\circ$
 $= -\frac{1}{\sqrt{3}}$

$\therefore$ eqn. of line is $y - 0 = -\frac{1}{\sqrt{3}}(x - 4)$

$\therefore \underline{\underline{y = -\frac{x}{\sqrt{3}} + \frac{4}{\sqrt{3}}}}$ (or $x + \sqrt{3}y - 4 = 0$)

b) i)



ii) $\begin{matrix} -2, 3 & 6, -1 \\ & \searrow \swarrow \\ & -1, 3 \end{matrix}$

$x = \frac{-6 - 6}{2}, y = \frac{9 + 1}{2}$

$\therefore \underline{\underline{P \text{ is } (-6, 5)}}$

c) i) $m_1 = 3, m_2 = -1$

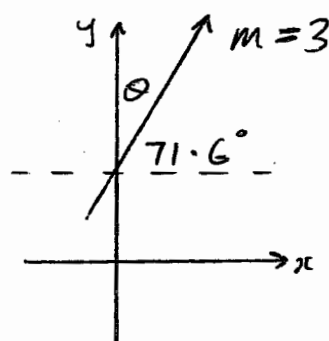
$\tan \theta = \left| \frac{3 - (-1)}{1 + 3(-1)} \right|$

$= \left| \frac{4}{-2} \right|$

$= 2$

$\therefore \underline{\underline{\theta = 63.4^\circ}}$

ii)



$\therefore \underline{\underline{\theta = 18.4^\circ}}$

5) a)

$P = (2, -1)$

$Q = (6, 5)$

grad. of PQ, $m_1 = \frac{6}{4} = \frac{3}{2}$

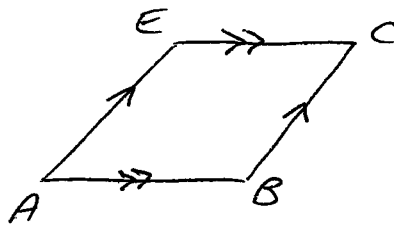
$\therefore m_2 = -\frac{2}{3}$ and use m.p. (4, 2)

$\therefore$ eqn. perp. bisector is $y - 2 = -\frac{2}{3}(x - 4)$

$3y - 6 = -2x + 8$

$\therefore \underline{\underline{2x + 3y - 14 = 0}}$

ii) E is $(-1, 5)$



b) i) $y = (4x-2)^{-\frac{1}{2}}$
 $\frac{dy}{dx} = -\frac{1}{2}(4x-2)^{-\frac{3}{2}} \times 4$
 $= \underline{\underline{-2(4x-2)^{-\frac{3}{2}}}}$
 (or equiv.)

ii) $\frac{dy}{dx} = \frac{2(4x-9)^4 - 4(4x-9)^3 \times 4 \times (2x+1)}{(4x-9)^8}$
 $= \frac{2(4x-9)^4 - 16(2x+1)(4x-9)^3}{(4x-9)^8}$
 $= \frac{2(4x-9) - 16(2x+1)}{(4x-9)^5}$
 $= \frac{8x - 18 - 32x - 16}{(4x-9)^5}$
 $= \underline{\underline{\frac{-24x - 34}{(4x-9)^5}}}$

c) $\lim_{x \rightarrow \infty} \frac{1 - \frac{5}{x} + \frac{2}{x^2}}{2 + \frac{3}{x}}$
 $= \underline{\underline{\frac{1}{2}}}$

⑥ a) First is gradient of secant (2 points)
 Second is gradient of curve, or tangent to curve (1 point)

b) $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h}$
 $= \lim_{h \rightarrow 0} \frac{x^2 - (x+h)^2}{x^2(x+h)^2} \times \frac{1}{h}$
 $= \lim_{h \rightarrow 0} \frac{\cancel{x^2} - \cancel{x^2} - 2xh - h^2}{x^2(x+h)^2} \times \frac{1}{h}$
 $= \lim_{h \rightarrow 0} \frac{\cancel{h}(-2x - h)}{x^2(x+h)^2} \times \frac{1}{\cancel{h}}$
 $= \frac{-2x}{x^4} = \underline{\underline{-\frac{2}{x^3}}}$

$$c) y = \frac{1}{2} x^{-1}$$

$$\begin{aligned} \frac{dy}{dx} &= -\frac{1}{2} x^{-2} \\ &= \frac{-1}{2x^2} \end{aligned}$$

$$\text{When } x=1, m_T = -\frac{1}{2}$$

$$m_N = 2$$

and point is $(1, \frac{1}{2})$

$$\therefore \text{eqn. of normal is } y - \frac{1}{2} = 2(x-1)$$

$$= 2x - 2$$

$$\therefore \underline{\underline{y = 2x - 1\frac{1}{2}}}$$

d) p.d. of $3x - 4y + k = 0$ from $(4, 0) < 2$ units (radius)

$$\therefore \frac{|3 \times 4 + 0 + k|}{\sqrt{3^2 + 4^2}} < 2$$

$$\frac{|12 + k|}{5} < 2$$

$$|12 + k| < 10$$

$$12 + k < 10 \text{ or } -12 - k < 10$$

$$k < -2 \text{ or } -k < 22$$

$$k > -22$$

$$\therefore \underline{\underline{-22 < k < -2}}$$